

NUWC_SDP17

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Chapter 1

Directory Hierarchy

1.1 Directories

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Chapter 2

Hierarchical Index

2.1 Class Hierarchy

This inheritance list is sorted roughly, but not completely, alphabetically:

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PinConfig	16
QObject	
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PdfReportGenerator	14
RecentFilesManager	17
SerialComms	19
TestController	23
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Chapter 3

Class Index

3.1 Class List

Here are the classes, structs, unions and interfaces with brief descriptions:

ChipConfiguration	Represents the configuration of a chip parsed from YAML	11
Exporter	Exports application data to JSON or CSV formats	12
PdfReportGenerator	Generates PDF reports summarizing test results	14
PinConfig	Represents configuration information for a single pin	16
RecentFilesManager	Handles persistence of recently opened files	17
SerialComms	Manages serial communication and data exchange with a microcontroller	19
TestController	Manages execution of tests using configuration data and serial communication	23
TestResult	Represents the result of a single test case	26
Tests	Represents a collection of test cases parsed from YAML	27
YamlProcessor	Handles loading and parsing of YAML files for chip configuration and tests	28

Chapter 4

File Index

4.1 File List

Here is a list of all files with brief descriptions:

/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/exporter.cpp	33
/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/exporter.h	
Class for exporting data in JSON & CSV formats	33
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Manages loading and saving of recently accessed files	36
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/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/yamlprocessor.cpp	40
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Parses YAML configuration files into structured data for testing	40

Chapter 5

Directory Documentation

5.1 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17 Directory Reference

Files

- file [exporter.cpp](#)
- file [exporter.h](#)
Class for exporting data in JSON & CSV formats.
- file [main.cpp](#)
- file [pdfreportgenerator.cpp](#)
- file [pdfreportgenerator.h](#)
Provides functionality for generating PDF test reports.
- file [recentfilesmanager.cpp](#)
- file [recentfilesmanager.h](#)
Manages loading and saving of recently accessed files.
- file [serialcomms.cpp](#)
- file [serialcomms.h](#)
Handles serial communication with an external MCU.
- file [testcontroller.cpp](#)
- file [testcontroller.h](#)
Controls execution of test sequences and coordinates system components.
- file [yamlprocessor.cpp](#)
- file [yamlprocessor.h](#)
Parses YAML configuration files into structured data for testing.

Chapter 6

Class Documentation

6.1 ChipConfiguration Struct Reference

Represents the configuration of a chip parsed from YAML.

```
#include <yamlprocessor.h>
```

Public Attributes

- QMap< QString, QString > [chipInfo](#)
- QMap< QString, QList< QString > > [pinNames](#)
- QMap< QString, QList< QString > > [pinConfigs](#)
- QMap< QString, QList< QString > > [rawPinConfigs](#)
- QSet< QString > [outputPins](#)
- QSet< QString > [clockPins](#)

6.1.1 Detailed Description

Represents the configuration of a chip parsed from YAML.

This structure contains metadata and pin configuration information required to initialize and interact with the chip under test.

6.1.2 Member Data Documentation

6.1.2.1 chipInfo

```
QMap<QString, QString> ChipConfiguration::chipInfo
```

General chip metadata (e.g., name, pin count)

6.1.2.2 clockPins

```
QSet<QString> ChipConfiguration::clockPins
```

Set of pins used as clock signals

6.1.2.3 outputPins

```
QSet<QString> ChipConfiguration::outputPins
```

Set of pins configured as outputs

6.1.2.4 pinConfigs

```
QMap<QString, QList<QString> > ChipConfiguration::pinConfigs
```

Processed pin configuration instructions

6.1.2.5 pinNames

```
QMap<QString, QList<QString> > ChipConfiguration::pinNames
```

Mapping of categories to pin names

6.1.2.6 rawPinConfigs

```
QMap<QString, QList<QString> > ChipConfiguration::rawPinConfigs
```

Raw pin configuration data from YAML

The documentation for this struct was generated from the following file:

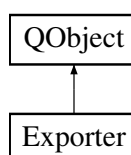
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/yamlprocessor.h

6.2 Exporter Class Reference

Exports application data to JSON or CSV formats.

```
#include <exporter.h>
```

Inheritance diagram for Exporter:



Public Member Functions

- [Exporter](#) ()
Constructs an [Exporter](#) object.
- Q_INVOKABLE void [exportData](#) (const QVariantMap &data, const QUrl &fileUrl, const QString &format)
Exports input data in specified format and stores in specified location.

6.2.1 Detailed Description

Exports application data to JSON or CSV formats.

This class formats input data and writes it to a specified file location.

6.2.2 Constructor & Destructor Documentation

6.2.2.1 Exporter()

```
Exporter::Exporter ()
```

Constructs an [Exporter](#) object.

6.2.3 Member Function Documentation

6.2.3.1 exportData()

```
void Exporter::exportData (  
    const QVariantMap & data,  
    const QUrl & fileUrl,  
    const QString & format)
```

Exports input data in specified format and stores in specified location.

This method is invocable from QML.

Parameters

<i>data</i>	The input data
<i>fileUrl</i>	The location that the formatted data should be stored
<i>format</i>	Output format. Supported values: "json", "csv".

Returns

None

The documentation for this class was generated from the following files:

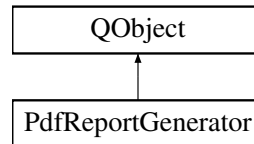
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[exporter.h](#)
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[exporter.cpp](#)

6.3 PdfReportGenerator Class Reference

Generates PDF reports summarizing test results.

```
#include <pdfreportgenerator.h>
```

Inheritance diagram for PdfReportGenerator:



Signals

- void [reportGenerated](#) (const QString &filePath)
Emitted when a report is successfully generated.
- void [reportFailed](#) (const QString &reason)
Emitted when report generation fails.

Public Member Functions

- [PdfReportGenerator](#) (QObject *parent=nullptr)
Constructs a PdfReportGenerator object.
- Q_INVOKABLE bool [generateBasicReport](#) (const QString &filePath, const QString &chipName, const QString &testMode, int totalTests, int passedTests, int failedTests, const QVariantList &pinConfigsVar, const QVariantList &testsVar, const QStringList &messages)
Generates a basic PDF report from test data.

6.3.1 Detailed Description

Generates PDF reports summarizing test results.

This class formats test results data and exports it as a PDF report. It supports integration with QML via `Q_INVOKABLE` methods and emits signals to indicate success or failure of report generation.

6.3.2 Constructor & Destructor Documentation

6.3.2.1 PdfReportGenerator()

```
PdfReportGenerator::PdfReportGenerator (
    QObject * parent = nullptr) [explicit]
```

Constructs a [PdfReportGenerator](#) object.

Parameters

<i>parent</i>	Optional parent QObject.
---------------	--------------------------

6.3.3 Member Function Documentation**6.3.3.1 generateBasicReport()**

```
bool PdfReportGenerator::generateBasicReport (
    const QString & filePath,
    const QString & chipName,
    const QString & testMode,
    int totalTests,
    int passedTests,
    int failedTests,
    const QVariantList & pinConfigsVar,
    const QVariantList & testsVar,
    const QStringList & messages)
```

Generates a basic PDF report from test data.

This method is invocable from QML via Q_INVOKABLE. It compiles test results, pin configurations, and summary data into a formatted PDF report saved to the specified file path.

Parameters

<i>filePath</i>	The destination file path for the generated PDF.
<i>chipName</i>	The name of the chip under test.
<i>testMode</i>	The mode in which testing was performed. Must be "Simulation" or "Hardware".
<i>totalTests</i>	Total number of tests executed.
<i>passedTests</i>	Number of tests that passed.
<i>failedTests</i>	Number of tests that failed.
<i>pinConfigsVar</i>	List of pin configuration objects (QVariantList).
<i>testsVar</i>	List of test result objects (QVariantList).
<i>messages</i>	Additional messages or notes to include in the report.

Returns

True if the report was generated successfully, false otherwise.

6.3.3.2 reportFailed

```
void PdfReportGenerator::reportFailed (
    const QString & reason) [signal]
```

Emitted when report generation fails.

Parameters

<i>reason</i>	Description of the failure.
---------------	-----------------------------

6.3.3.3 reportGenerated

```
void PdfReportGenerator::reportGenerated (  
    const QString & filePath) [signal]
```

Emitted when a report is successfully generated.

Parameters

<i>filePath</i>	The file path of the generated report.
-----------------	--

The documentation for this class was generated from the following files:

- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[pdfreportgenerator.h](#)
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[pdfreportgenerator.cpp](#)

6.4 PinConfig Struct Reference

Represents configuration information for a single pin.

```
#include <pdfreportgenerator.h>
```

Public Attributes

- QString [pinName](#)
- QString [direction](#)
- QString [defaultValue](#)

6.4.1 Detailed Description

Represents configuration information for a single pin.

This structure stores metadata about a pin, including its name, direction, and default value.

6.4.2 Member Data Documentation

6.4.2.1 defaultValue

```
QString PinConfig::defaultValue
```

Default value assigned to the pin

6.4.2.2 direction

```
QString PinConfig::direction
```

Direction of the pin ("Input" or "Output")

6.4.2.3 pinName

```
QString PinConfig::pinName
```

Name of the pin

The documentation for this struct was generated from the following file:

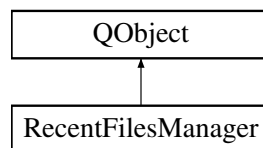
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[pdfreportgenerator.h](#)

6.5 RecentFileManager Class Reference

Handles persistence of recently opened files.

```
#include <recentfilesmanager.h>
```

Inheritance diagram for RecentFileManager:



Public Member Functions

- [RecentFileManager](#) (QObject *parent=nullptr)
Constructs a [RecentFileManager](#) object.
- Q_INVOKABLE QVariantList [loadRecentFiles](#) ()
Loads the list of recent files.
- Q_INVOKABLE void [saveRecentFiles](#) (const QVariantList &files)
Saves the list of recent files.

6.5.1 Detailed Description

Handles persistence of recently opened files.

This class provides functionality to load and save a list of recently accessed files. It is designed to be used from QML via Q_INVOKABLE methods.

6.5.2 Constructor & Destructor Documentation

6.5.2.1 RecentFileManager()

```
RecentFileManager::RecentFileManager (  
    QObject * parent = nullptr) [explicit]
```

Constructs a [RecentFileManager](#) object.

Parameters

<i>parent</i>	Optional parent QObject.
---------------	--------------------------

6.5.3 Member Function Documentation

6.5.3.1 loadRecentFiles()

```
QVariantList RecentFileManager::loadRecentFiles ()
```

Loads the list of recent files.

This method retrieves the stored list of recently accessed files.

Returns

A QVariantList containing recent file entries.

6.5.3.2 saveRecentFiles()

```
void RecentFileManager::saveRecentFiles (  
    const QVariantList & files)
```

Saves the list of recent files.

This method persists the provided list of recent file entries.

Parameters

<i>files</i>	A QVariantList containing file entries to store.
--------------	--

The documentation for this class was generated from the following files:

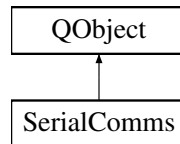
- [/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/recentfilesmanager.h](#)
- [/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/recentfilesmanager.cpp](#)

6.6 SerialComms Class Reference

Manages serial communication and data exchange with a microcontroller.

```
#include <serialcomms.h>
```

Inheritance diagram for SerialComms:



Signals

- void [dataReceived](#) (QString message)
Emitted when data is received from the MCU.

Public Member Functions

- [SerialComms](#) (QObject *parent=nullptr)
Constructs a [SerialComms](#) object.
- [~SerialComms](#) ()
Destroys the [SerialComms](#) object.
- void [sendByteStream](#) (QByteArray byteStream, bool useCRC)
Sends a byte stream to the MCU.
- QString [readMCU](#) (bool useCRC=false, bool testArduino=true)
Reads data from the MCU.
- uint16_t [calculateCRC](#) (const QByteArray &data)
Calculates a CRC checksum for the given data.
- bool [verifyCRC](#) (const QByteArray &data)
Verifies the CRC of received data.
- void [listAvailablePorts](#) ()
Lists available serial ports.
- Q_INVOKABLE QString [sendTestStream](#) (QString stream)
Sends a test stream to the MCU.
- Q_INVOKABLE bool [isMCUConnected](#) ()
Checks whether the MCU is currently connected.

6.6.1 Detailed Description

Manages serial communication and data exchange with a microcontroller.

This class provides functionality to send and receive data over a serial connection. It supports optional CRC-based integrity checking and exposes certain methods to QML for interaction with the GUI.

6.6.2 Constructor & Destructor Documentation

6.6.2.1 SerialComms()

```
SerialComms::SerialComms (  
    QObject * parent = nullptr) [explicit]
```

Constructs a [SerialComms](#) object.

Parameters

<i>parent</i>	Optional parent QObject.
---------------	--------------------------

6.6.2.2 ~SerialComms()

```
SerialComms::~~SerialComms ()
```

Destroys the [SerialComms](#) object.

6.6.3 Member Function Documentation

6.6.3.1 calculateCRC()

```
uint16_t SerialComms::calculateCRC (  
    const QByteArray & data)
```

Calculates a CRC checksum for the given data.

Parameters

<i>data</i>	Input byte array.
-------------	-------------------

Returns

The computed CRC value.

6.6.3.2 dataReceived

```
void SerialComms::dataReceived (  
    QString message) [signal]
```

Emitted when data is received from the MCU.

Parameters

<i>message</i>	The received message.
----------------	-----------------------

6.6.3.3 isMCUConnected()

```
bool SerialComms::isMCUConnected ()
```

Checks whether the MCU is currently connected.

This method is invokable from QML.

Returns

True if the MCU is connected, false otherwise.

6.6.3.4 listAvailablePorts()

```
void SerialComms::listAvailablePorts ()
```

Lists available serial ports.

Primarily used for debugging connection issues.

6.6.3.5 readMCU()

```
QString SerialComms::readMCU (
    bool useCRC = false,
    bool testArduino = true)
```

Reads data from the MCU.

Optionally verifies CRC and supports a test mode for debugging.

Parameters

<i>useCRC</i>	Whether to validate CRC on received data.
<i>testArduino</i>	Whether to use test mode behavior.

Returns

The received message as a QString.

6.6.3.6 sendByteStream()

```
void SerialComms::sendByteStream (
    QByteArray byteStream,
    bool useCRC)
```

Sends a byte stream to the MCU.

Optionally appends a CRC checksum before transmission.

Parameters

<i>byteStream</i>	The raw data to send.
<i>useCRC</i>	Whether to append a CRC checksum.

6.6.3.7 sendTestStream()

```
QString SerialComms::sendTestStream (
    QString stream)
```

Sends a test stream to the MCU.

This method is invocable from QML and is intended for debugging.

Parameters

<i>stream</i>	The test data to send.
---------------	------------------------

Returns

The MCU response.

6.6.3.8 verifyCRC()

```
bool SerialComms::verifyCRC (
    const QByteArray & data)
```

Verifies the CRC of received data.

Parameters

<i>data</i>	Input byte array including checksum.
-------------	--------------------------------------

Returns

True if CRC is valid, false otherwise.

The documentation for this class was generated from the following files:

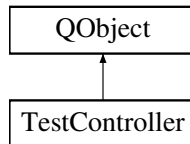
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[serialcomms.h](#)
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[serialcomms.cpp](#)

6.7 TestController Class Reference

Manages execution of tests using configuration data and serial communication.

```
#include <testcontroller.h>
```

Inheritance diagram for TestController:



Signals

- void [logMessage](#) (QString text, QString color)
Emits a log message for UI display.
- void [resultsReady](#) (QVariantMap results)
Emitted when test results are ready.
- void [clearConsole](#) ()
Clears the console/log output in the UI.
- void [testCompleted](#) ()
Emitted when all tests have completed.

Public Member Functions

- [TestController](#) (YamlProcessor *yamlProcessor, [SerialComms](#) *serialComms, QObject *parent=nullptr)
Constructs a [TestController](#) object.
- Q_INVOKABLE void [startTests](#) (const QUrl &filePath, bool isSimulation)
Starts execution of tests from a YAML file.
- Q_INVOKABLE void [pauseTests](#) ()
Pauses the currently running test sequence.
- Q_INVOKABLE void [resumeTests](#) ()
Resumes a paused test sequence.
- Q_INVOKABLE void [stopTests](#) ()
Stops the currently running test sequence.

6.7.1 Detailed Description

Manages execution of tests using configuration data and serial communication.

This class coordinates the process of running tests defined in a YAML file. It interacts with the [YamlProcessor](#) to parse test configurations and uses [SerialComms](#) to communicate with the MCU. It supports stepwise execution, pausing, resuming, and stopping of tests, and emits signals for UI updates.

6.7.2 Constructor & Destructor Documentation

6.7.2.1 TestController()

```
TestController::TestController (
    YamlProcessor * yamlProcessor,
    SerialComms * serialComms,
    QObject * parent = nullptr) [explicit]
```

Constructs a [TestController](#) object.

Parameters

<i>yamlProcessor</i>	Pointer to the YAML processing component.
<i>serialComms</i>	Pointer to the serial communication handler.
<i>parent</i>	Optional parent QObject.

6.7.3 Member Function Documentation

6.7.3.1 clearConsole

```
void TestController::clearConsole () [signal]
```

Clears the console/log output in the UI.

6.7.3.2 logMessage

```
void TestController::logMessage (
    QString text,
    QString color) [signal]
```

Emits a log message for UI display.

Parameters

<i>text</i>	The message content.
<i>color</i>	Display color (e.g., for status indication).

6.7.3.3 pauseTests()

```
void TestController::pauseTests ()
```

Pauses the currently running test sequence.

This method is invocable from QML.

6.7.3.4 resultsReady

```
void TestController::resultsReady (
    QVariantMap results) [signal]
```

Emitted when test results are ready.

Parameters

<i>results</i>	Aggregated test results.
----------------	--------------------------

6.7.3.5 resumeTests()

```
void TestController::resumeTests ()
```

Resumes a paused test sequence.

This method is invocable from QML.

6.7.3.6 startTests()

```
void TestController::startTests (
    const QUrl & filePath,
    bool isSimulation)
```

Starts execution of tests from a YAML file.

This method is invocable from QML and initializes the test run.

Parameters

<i>filePath</i>	Path to the YAML test configuration file.
<i>isSimulation</i>	Whether to run in simulation mode or hardware mode.

6.7.3.7 stopTests()

```
void TestController::stopTests ()
```

Stops the currently running test sequence.

This method is invocable from QML.

6.7.3.8 testCompleted

```
void TestController::testCompleted () [signal]
```

Emitted when all tests have completed.

The documentation for this class was generated from the following files:

- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[testcontroller.h](#)
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[testcontroller.cpp](#)

6.8 TestResult Struct Reference

Represents the result of a single test case.

```
#include <pdfreportgenerator.h>
```

Public Attributes

- QString [testName](#)
- QMap< QString, QString > [inputPins](#)
- QString [responseBytes](#)
- QString [expectedBytes](#)
- bool [passed](#)
- double [durationMs](#)

6.8.1 Detailed Description

Represents the result of a single test case.

This structure contains all relevant data for a test execution, including input values, expected and actual outputs, and status.

6.8.2 Member Data Documentation

6.8.2.1 durationMs

```
double TestResult::durationMs
```

Execution time in milliseconds

6.8.2.2 expectedBytes

```
QString TestResult::expectedBytes
```

Expected response

6.8.2.3 inputPins

```
QMap<QString, QString> TestResult::inputPins
```

Mapping of pin names to input values

6.8.2.4 passed

```
bool TestResult::passed
```

Indicates whether the test passed

6.8.2.5 responseBytes

```
QString TestResult::responseBytes
```

Actual response from the test

6.8.2.6 testName

```
QString TestResult::testName
```

Name of the test

The documentation for this struct was generated from the following file:

- [/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/pdfreportgenerator.h](#)

6.9 Tests Struct Reference

Represents a collection of test cases parsed from YAML.

```
#include <yamlprocessor.h>
```

Public Attributes

- `QList< QList< QPair< QString, QList< QString > > > > tests`
- `QList< QList< QString > > outputs`
- `QList< QString > descriptions`
- `QList< QMap< int, QString > > pinValueMaps`

6.9.1 Detailed Description

Represents a collection of test cases parsed from YAML.

This structure stores test instructions, expected outputs, and metadata required for executing and validating tests.

6.9.2 Member Data Documentation

6.9.2.1 descriptions

```
QList<QString> Tests::descriptions
```

Descriptions for each test case

6.9.2.2 outputs

```
QList<QList<QString> > Tests::outputs
```

Expected outputs for each test

6.9.2.3 pinValueMaps

```
QList<QMap<int, QString> > Tests::pinValueMaps
```

Mapping of pin indices to values

6.9.2.4 tests

```
QList<QList<QPair<QString, QList<QString> > > > Tests::tests
```

Test instructions (command + pin values)

The documentation for this struct was generated from the following file:

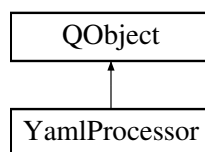
- [/Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/yamlprocessor.h](#)

6.10 YamlProcessor Class Reference

Handles loading and parsing of YAML files for chip configuration and tests.

```
#include <yamlprocessor.h>
```

Inheritance diagram for YamlProcessor:



Public Slots

- [ChipConfiguration readChipConfiguration](#) (const [QUrl](#) &filePath)
Reads chip configuration from a YAML file.
- [Tests readTests](#) (const [QUrl](#) &filePath, const [ChipConfiguration](#) &cfg)
Reads test definitions from a YAML file.

Signals

- void [yamlLoaded](#) (const [QUrl](#) &filePath)
Emitted when a YAML file has been successfully loaded.

Public Member Functions

- [YamlProcessor](#) (QObject *parent=nullptr)
Constructs a [YamlProcessor](#) object.
- Q_INVOKABLE QVariantMap [loadYaml](#) (const QString &filePath)
Loads a YAML file and returns its contents as a QVariantMap.

6.10.1 Detailed Description

Handles loading and parsing of YAML files for chip configuration and tests.

This class provides functionality to read YAML files and convert them into structured data used by the [TestController](#). It exposes methods for QML usage and emits signals when YAML files are successfully loaded.

6.10.2 Constructor & Destructor Documentation

6.10.2.1 YamlProcessor()

```
YamlProcessor::YamlProcessor (  
    QObject * parent = nullptr) [explicit]
```

Constructs a [YamlProcessor](#) object.

Parameters

<i>parent</i>	Optional parent QObject.
---------------	--------------------------

6.10.3 Member Function Documentation

6.10.3.1 loadYaml()

```
QVariantMap YamlProcessor::loadYaml (  
    const QString & filePath)
```

Loads a YAML file and returns its contents as a QVariantMap.

This method is invocable from QML and provides a generic representation of the YAML structure.

Parameters

<i>filePath</i>	Path to the YAML file.
-----------------	------------------------

Returns

Parsed YAML data as a QVariantMap.

High-level wrapper that loads YAML and converts it into a QVariantMap for easy use in QML/UI layers.

6.10.3.2 readChipConfiguration

```
ChipConfiguration YamlProcessor::readChipConfiguration (  
    const QUrl & filePath) [slot]
```

Reads chip configuration from a YAML file.

Parses metadata and pin configuration required to initialize the chip.

Parameters

<i>filePath</i>	Path to the YAML file.
-----------------	------------------------

Returns

A [ChipConfiguration](#) struct containing parsed data.

Parses chip configuration data from a YAML file. Extracts chip metadata, pin mappings, and configuration groupings.

6.10.3.3 readTests

```
Tests YamlProcessor::readTests (  
    const QUrl & filePath,  
    const ChipConfiguration & cfg) [slot]
```

Reads test definitions from a YAML file.

Uses the provided chip configuration to correctly interpret test instructions and expected outputs.

Parameters

<i>filePath</i>	Path to the YAML file.
<i>cfg</i>	Parsed chip configuration.

Returns

A [Tests](#) struct containing parsed test data.

Parses test cases from YAML and converts them into structured instructions. Handles sequential logic, pin expansion, and expected output generation.

6.10.3.4 yamlLoaded

```
void YamlProcessor::yamlLoaded (
    const QUrl & filePath) [signal]
```

Emitted when a YAML file has been successfully loaded.

Parameters

<i>filePath</i>	Path to the loaded YAML file.
-----------------	-------------------------------

The documentation for this class was generated from the following files:

- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[yamlprocessor.h](#)
- /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/[yamlprocessor.cpp](#)

Chapter 7

File Documentation

7.1 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ exporter.cpp File Reference

```
#include "exporter.h"  
#include <QFile>  
#include <QJsonDocument>  
#include <QJsonObject>  
#include <QJsonArray>  
#include <QTextStream>  
#include <QFileDialog>  
#include <QDir>
```

7.2 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ exporter.h File Reference

Class for exporting data in JSON & CSV formats.

```
#include <QObject>
```

Classes

- class [Exporter](#)
Exports application data to JSON or CSV formats.

7.2.1 Detailed Description

Class for exporting data in JSON & CSV formats.

7.3 exporter.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 #ifndef EXPORTER_H
00007 #define EXPORTER_H
00008
00009 #include <QObject>
00010
00018 class Exporter : public QObject
00019 {
00020     Q_OBJECT
00021 public:
00025     Exporter();
00026
00038     Q_INVOKABLE void exportData(const QVariantMap &data, const QUrl &fileUrl, const QString &format);
00039 };
00040
00041 #endif // EXPORTER_H
```

7.4 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ main.cpp File Reference

```
#include <QGuiApplication>
#include <QQmlApplicationEngine>
#include <QQmlContext>
#include <QQuickStyle>
#include "yamlprocessor.h"
#include "serialcomms.h"
#include "testcontroller.h"
#include "recentfilesmanager.h"
#include "exporter.h"
#include "pdfreportgenerator.h"
```

Functions

- int [main](#) (int argc, char *argv[])

7.4.1 Function Documentation

7.4.1.1 main()

```
int main (
    int argc,
    char * argv[])
```

7.5 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/pdfreportgenerator.cpp File Reference

```
#include "pdfreportgenerator.h"
#include <QPdfWriter>
#include <QPainter>
#include <QPageSize>
#include <QMarginsF>
#include <QFileInfo>
#include <QDir>
#include <QDateTime>
#include <QColor>
#include <QFont>
#include <QRect>
#include <QStringList>
```

7.6 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/pdfreportgenerator.h File Reference

Provides functionality for generating PDF test reports.

```
#include <QObject>
#include <QStringList>
#include <QMap>
```

Classes

- struct [PinConfig](#)
Represents configuration information for a single pin.
- struct [TestResult](#)
Represents the result of a single test case.
- class [PdfReportGenerator](#)
Generates PDF reports summarizing test results.

7.6.1 Detailed Description

Provides functionality for generating PDF test reports.

7.7 pdfreportgenerator.h

[Go to the documentation of this file.](#)

```

00001
00005
00006 #ifndef PDFREPORTGENERATOR_H
00007 #define PDFREPORTGENERATOR_H
00008
00009 #include <QObject>
00010 #include <QStringList>
00011 #include <QMap>
00012
00020 struct PinConfig {
00021     QString pinName;
00022     QString direction;
00023     QString defaultValue;
00024 };
00025
00033 struct TestResult {
00034     QString testName;
00035
00036     QMap<QString, QString> inputPins;
00037
00038     QString responseBytes;
00039     QString expectedBytes;
00040
00041     bool passed;
00042     double durationMs;
00043 };
00044
00053 class PdfReportGenerator : public QObject
00054 {
00055     Q_OBJECT
00056
00057 public:
00062     explicit PdfReportGenerator(QObject *parent = nullptr);
00063
00083     Q_INVOKABLE bool generateBasicReport(const QString &filePath,
00084                                         const QString &chipName,
00085                                         const QString &testMode,
00086                                         int totalTests,
00087                                         int passedTests,
00088                                         int failedTests,
00089                                         const QVariantList &pinConfigsVar,
00090                                         const QVariantList &testsVar,
00091                                         const QStringList &messages);
00092
00093 signals:
00098     void reportGenerated(const QString &filePath);
00099
00104     void reportFailed(const QString &reason);
00105 };
00106
00107 #endif // PDFREPORTGENERATOR_H

```

7.8 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ recentfilesmanager.cpp File Reference

```

#include "recentfilesmanager.h"
#include <QFile>
#include <QJsonDocument>
#include <QJsonArray>
#include <QStandardPaths>
#include <QDir>

```

7.9 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ recentfilesmanager.h File Reference

Manages loading and saving of recently accessed files.

```
#include <QObject>
#include <QVariantList>
```

Classes

- class [RecentFilesManager](#)
Handles persistence of recently opened files.

7.9.1 Detailed Description

Manages loading and saving of recently accessed files.

7.10 recentfilesmanager.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 #ifndef RECENTFILESMANAGER_H
00007 #define RECENTFILESMANAGER_H
00008
00009 #include <QObject>
00010 #include <QVariantList>
00011
00020 class RecentFilesManager : public QObject {
00021     Q_OBJECT
00022
00023 public:
00028     explicit RecentFilesManager(QObject *parent = nullptr);
00029
00037     Q_INVOKABLE QVariantList loadRecentFiles();
00038
00046     Q_INVOKABLE void saveRecentFiles(const QVariantList &files);
00047
00048 private:
00057     QString getFilePath();
00058 };
00059
00060 #endif // RECENTFILESMANAGER_H
```

7.11 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/serialcomms.cpp File Reference

```
#include "serialcomms.h"
#include <QDebug>
#include <QSerialPort>
#include <QSerialPortInfo>
#include <QCoreApplication>
#include <QByteArray>
#include <QStringList>
#include <QThread>
#include <QObject>
#include <QTimer>
#include <QMap>
#include <QtCore/qvariant.h>
```

7.12 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/serialcomms.h File Reference

Handles serial communication with an external MCU.

```
#include <QObject>
#include <QSerialPort>
#include <QSerialPortInfo>
#include <QThread>
#include <QByteArray>
```

Classes

- class [SerialComms](#)

Manages serial communication and data exchange with a microcontroller.

7.12.1 Detailed Description

Handles serial communication with an external MCU.

7.13 serialcomms.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 #ifndef SERIALCOMMS_H
00007 #define SERIALCOMMS_H
00008
00009 #include <QObject>
00010 #include <QSerialPort>
00011 #include <QSerialPortInfo>
00012 #include <QThread>
00013 #include <QByteArray>
00014
00023 class SerialComms : public QObject
00024 {
00025     Q_OBJECT
00026
00027 public:
00032     explicit SerialComms(QObject *parent = nullptr);
00033
00037     ~SerialComms();
00038
00047     void sendByteStream(QByteArray byteStream, bool useCRC);
00048
00058     QString readMCU(bool useCRC = false, bool testArduino = true);
00059
00066     uint16_t calculateCRC(const QByteArray &data);
00067
00074     bool verifyCRC(const QByteArray &data);
00075
00081     void listAvailablePorts();
00082
00091     Q_INVOKABLE QString sendTestStream(QString stream);
00092
00100     Q_INVOKABLE bool isMCUConnected();
00101
00102 signals:
00107     void dataReceived(QString message);
00108
00109 private:
00110     QSerialPort *serialPort = nullptr;
00111     QByteArray msgBuffer;
00112 };
00113
00114 #endif // SERIALCOMMS_H
```

7.14 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ testcontroller.cpp File Reference

```
#include "testcontroller.h"
```

7.15 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/↵ testcontroller.h File Reference

Controls execution of test sequences and coordinates system components.

```
#include <QObject>
#include <yamlprocessor.h>
#include <serialcomms.h>
#include <QTimer>
#include <QElapsedTimer>
```

Classes

- class [TestController](#)

Manages execution of tests using configuration data and serial communication.

7.15.1 Detailed Description

Controls execution of test sequences and coordinates system components.

7.16 testcontroller.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 #ifndef TESTCONTROLLER_H
00007 #define TESTCONTROLLER_H
00008
00009 #include <QObject>
00010 #include <yamlprocessor.h>
00011 #include <serialcomms.h>
00012 #include <QTimer>
00013 #include <QElapsedTimer>
00014
00024 class TestController : public QObject
00025 {
00026     Q_OBJECT
00027
00028 public:
00036     explicit TestController(YamlProcessor* yamlProcessor,
00037                             SerialComms* serialComms,
00038                             QObject* parent = nullptr);
00039
00048     Q_INVOKABLE void startTests(const QUrl &filePath, bool isSimulation);
00049
00055     Q_INVOKABLE void pauseTests();
00056
00062     Q_INVOKABLE void resumeTests();
00063
```

```

00069     Q_INVOKABLE void stopTests();
00070
00071 private:
00072     void sendChipConfiguration(const ChipConfiguration& config);
00073
00074     void resolveSequentialOutputs(
00075         const QList<QPair<QString, QList<QString>>& test,
00076         QList<QString>& expectedOutputs,
00077         int pinCount);
00078
00079     void sendTest(const QList<QPair<QString, QList<QString>>& test, int pinCount);
00080
00081     QByteArray createExpectedBytestream(QList<QString>& expectedOutputs);
00082
00083     void runTestsStepwise(const QUrl &filePath, bool isSimulation);
00084
00085     QVariantMap results;
00086     int currentTestIndex;
00087     Tests currentTests;
00088     ChipConfiguration currentConfig;
00089     QUrl currentFile;
00090     bool currentSimulation;
00091
00092     bool isRunning = false;
00093     bool isPaused = false;
00094     bool isStopped = true;
00095
00096     YamlProcessor* yamlProcessor;
00097     SerialComms* serialComms;
00098
00099     QElapsedTimer testTimer;
00100
00101 signals:
00102     void logMessage(QString text, QString color);
00103
00104     void resultsReady(QVariantMap results);
00105
00106     void clearConsole();
00107
00108     void testCompleted();
00109 };
00110
00111 #endif // TESTCONTROLLER_H

```

7.17 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/ ↩ yamlprocessor.cpp File Reference

```

#include "yamlprocessor.h"
#include <QDebug>
#include <QFile>
#include <QFileInfo>
#include <QMap>
#include <QString>
#include <map>
#include <QList>
#include <QVariantMap>
#include <QVariantList>
#include <yaml-cpp/yaml.h>

```

7.18 /Users/arnav/Documents/GitHub/NUWC-SDP17/NUWC_SDP17/ ↩ yamlprocessor.h File Reference

Parses YAML configuration files into structured data for testing.

```

#include <QObject>
#include <QUrl>

```

```
#include <QString>
#include <QMap>
#include <QList>
#include <QVariantMap>
#include <QSet>
```

Classes

- struct [ChipConfiguration](#)
Represents the configuration of a chip parsed from YAML.
- struct [Tests](#)
Represents a collection of test cases parsed from YAML.
- class [YamlProcessor](#)
Handles loading and parsing of YAML files for chip configuration and tests.

7.18.1 Detailed Description

Parses YAML configuration files into structured data for testing.

7.19 yamlprocessor.h

[Go to the documentation of this file.](#)

```
00001
00005
00006 // Code Author: Arnav Thakrar
00007
00008 #ifndef YAMLPROCESSOR_H
00009 #define YAMLPROCESSOR_H
00010
00011 #include <QObject>
00012 #include <QUrl>
00013 #include <QString>
00014 #include <QMap>
00015 #include <QList>
00016 #include <QVariantMap>
00017 #include <QSet>
00018
00026 struct ChipConfiguration {
00027     QMap<QString, QString> chipInfo;
00028     QMap<QString, QList<QString>> pinNames;
00029     QMap<QString, QList<QString>> pinConfigs;
00030     QMap<QString, QList<QString>> rawPinConfigs;
00031     QSet<QString> outputPins;
00032     QSet<QString> clockPins;
00033 };
00034
00042 struct Tests {
00043     QList<QList<QPair<QString, QList<QString>>>> tests;
00044     QList<QList<QString>> outputs;
00045     QList<QString> descriptions;
00046     QList<QMap<int, QString>> pinValueMaps;
00047 };
00048
00057 class YamlProcessor : public QObject
00058 {
00059     Q_OBJECT
00060
00061 public:
00066     explicit YamlProcessor(QObject *parent = nullptr);
00067
00077     Q_INVOKABLE QVariantMap loadYaml(const QString &filePath);
00078
00079 public slots:
00088     ChipConfiguration readChipConfiguration(const QUrl& filePath);
00089
```

```
00100     Tests readTests(const QUrl& filePath, const ChipConfiguration& cfg);
00101
00102 signals:
00107     void yamlLoaded(const QUrl& filePath);
00108 };
00109
00110 #endif // YAMLPROCESSOR_H
```

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